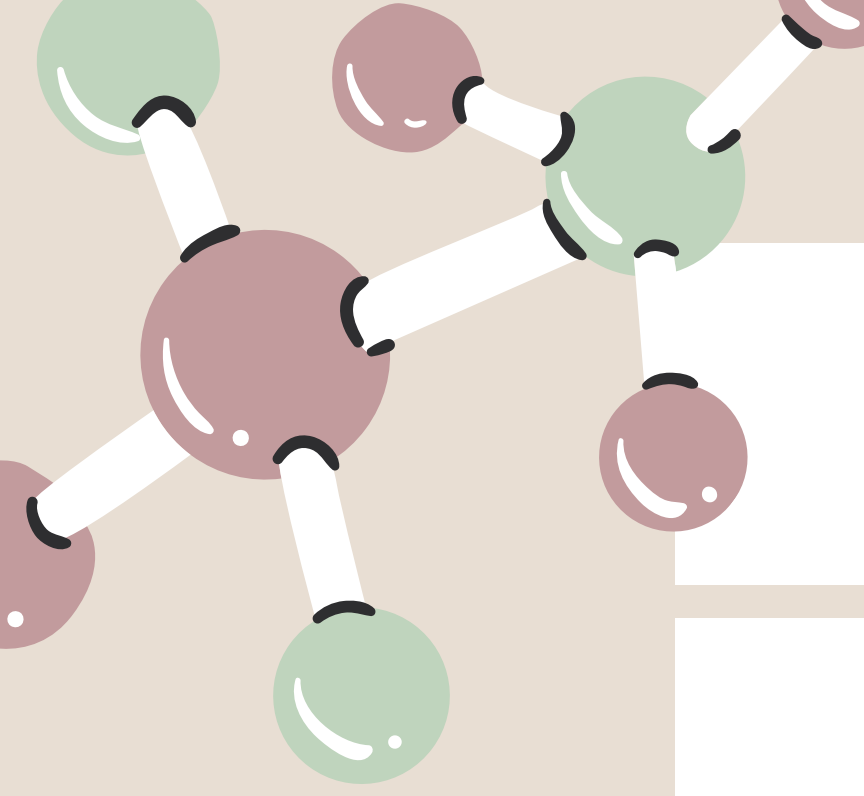
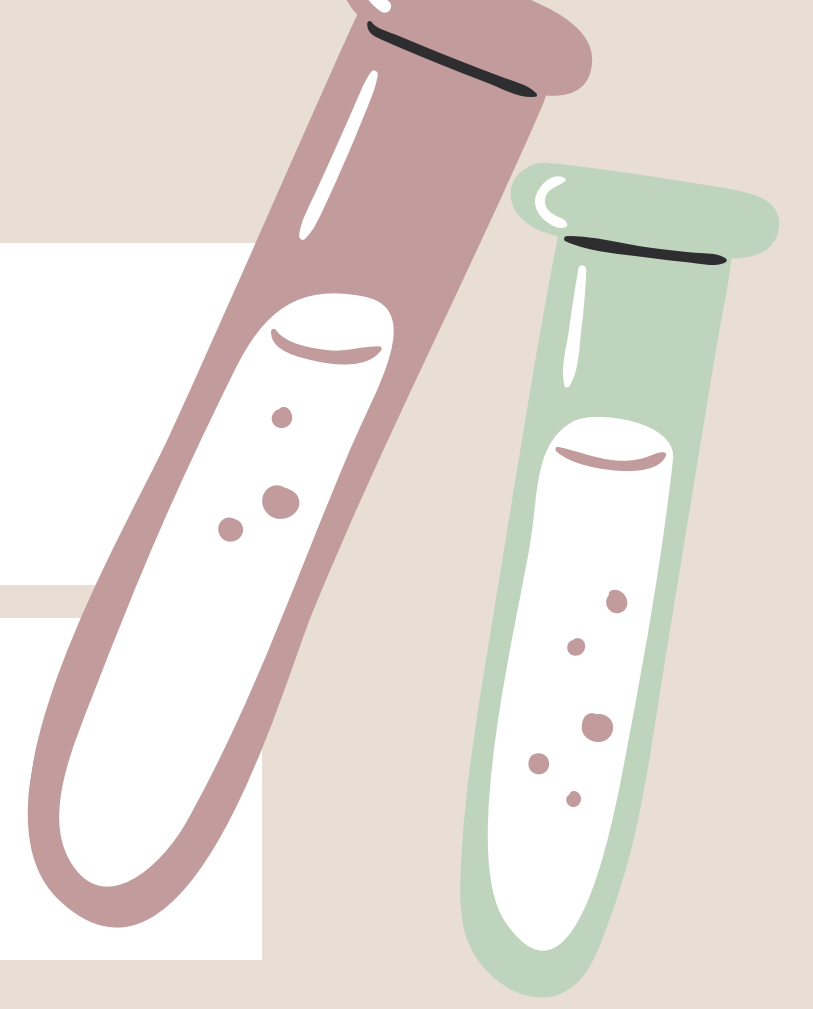




DNA Extraction



LESSON

This lab allows students to visualize DNA, the most important nucleic acid in the body that determines appearance, disease, and heredity. The lab demonstrates concepts of chemical and mechanical lysis (mashing the bananas and adding dish soap), denaturation (when DNA unwinds itself from other proteins when salt is added), and solubility (the DNA doesn't dissolve in alcohol, instead, it clumps together). All of these features of DNA have important functions that protect them inside the human body.

EXPERIMENT

A cup is filled with salt and water. The banana is completely mashed inside the bag, then the saltwater is added. The mixture is squeezed together for 30-45 seconds. The dishwashing liquid is then mixed in without creating too much foam. The coffee filter is secured over the cup, and the mixture is poured inside until all the liquid is in the cup. After the filter is removed, alcohol is slowly added down the side of the cup until a 1-2 in thick layer has been formed. After waiting for about 8 minutes, bubbles and cloudy material may form. Using the stirrer to separate the cloudy substance, DNA has been collected!

Materials

- 1/2 ripe banana
- 1/2 cup hot water
- 1 tsp salt
- 1/2 tsp dishwashing liquid
- Resealable bag
- Cold isopropyl alcohol
- Coffee filter
- Stirrer
- Glass/cup

